

Critical Decision Method - CDM Eliciting Knowledge

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CRITICAL DECISION METHOD

In 1988, Gary and colleagues Roberta Calderwood and Anne Clinton-Cirocco developed a new method of knowledge elicitation, the Critical Decision Method (CDM), as part of a project with fire ground commanders for the U.S. Army Research Institute.

The CDM is a retrospective interview strategy that applies a set of cognitive probes to actual, non-routine events that require expert decision-making. Once an incident is selected, the interviewer uses a semi-structured format to probe different aspects of the decision making process.

The CDM has been used widely to elicit expertise in domains including healthcare, military, emergency response, and more.

For additional reading on the CDM, click here.

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CDM´s reference

1. Klein G a, Calderwood R, Macgregor D. Critical Decision Method for Eliciting Knowledge. Ieee Trans Syst Man Cybern [Internet]. 1989;19(3):462–72. Available from: http://ieeexplore.ieee.org/xpls/abs_all.jsp?arnumber=31053

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RPD´s Klein Model

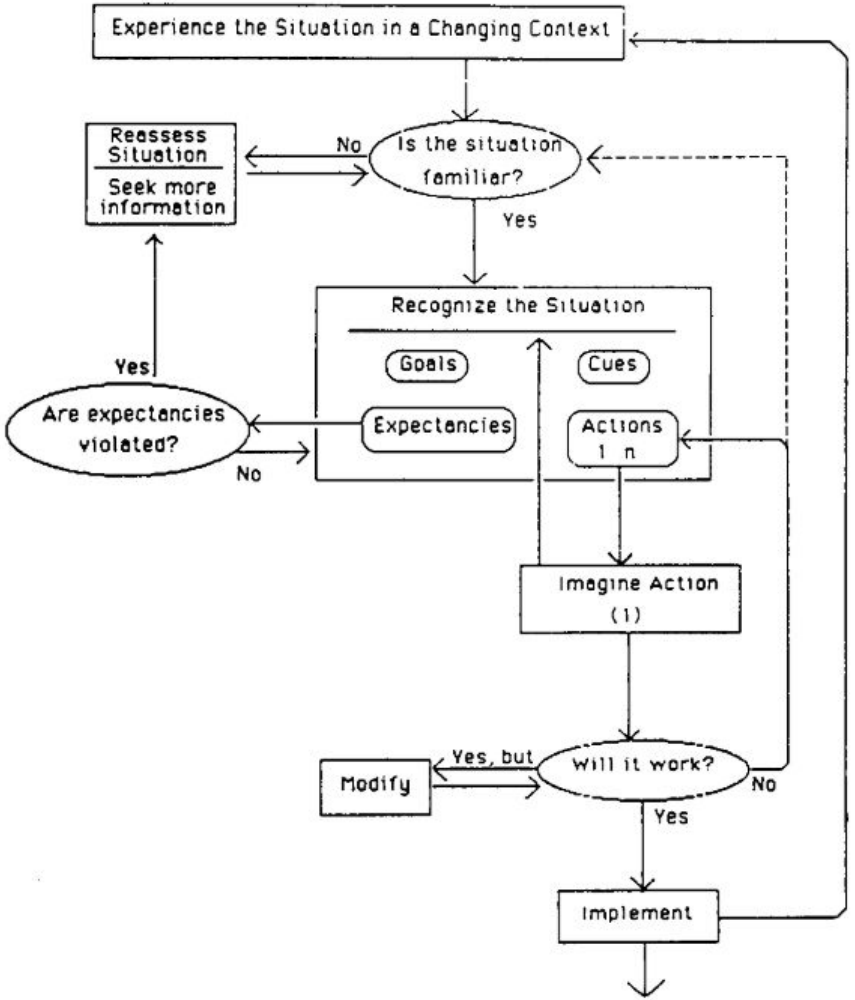


Fig. 1. Recognition-primed decision (RPD) model.

CDM´s Method

The CDM is a retrospective interview strategy that applies a set of cognitive probes to actual nonroutine incidents that required expert judgment or decisionmaking.

> Once the incident is selected, the interviewer asks for a brief description. Then a semistructured format is used to probe different aspects of the decisionmaking process. Specific procedures have also been developed for analyzing the data.

CDM´s Method

Focus: nonroutine cases.

The focus on nonroutine cases also makes the method most appropriate to eliciting tacit knowledge that is not part of the formalized procedures for a domain.

CDM´s Method

Procedures

Core Procedures

The procedures adopted for the critical decision interviews represent our solutions to meeting these goals and practical considerations. The basic procedure can be summarized in the following steps.

Step 1 — Select Incident: Incidents are selected that can illustrate nonroutine aspects of a domain. The idea is to

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challenge for the individual. Thus we ask the decision-maker to select an incident that was challenging and that, in his or her decisionmaking, might have differed from someone with less experience. This selection criterion is common to critical methods (e.g., [22]).

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Step 2—Obtain Unstructured Incident Account: The officer was asked to describe the incident from the time he received the alarm to the time when the incident was judged to be under control. For the most part this account proceeded without interruption by the interviewers, except for minor points of clarification. The procedure accomplished several goals. First, it created a context for understanding on the part of the interviewer. It was important to us to have a sense of the individual's phenomenological perspective of the event as a bulwark against our own biases and the contamination of the questioning procedure itself. Second, the account served to activate the officer's memory of the event as a context for questioning. In addition, we judged that the procedure helped us achieve a high level of cooperation from the officers by establishing us as listeners rather than interrogators. Obtaining cooper-

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Step 3—Construct Incident Timeline: After the incident had been related, the interviewer proceeded to reconstruct the account in the form of a timeline that established the sequence and duration of each event reported by the officer. Events included both objectively verifiable occurrences (e.g., "the second alarm equipment arrived two minutes later") and thoughts and perceptions reported by the officer (e.g., "the color of the smoke indicated the presence of a toxic substance. I thought I might have to call a second alarm at this point"). The timeline served to establish a shared awareness of the "facts of the case" from the officer's perspective. Many times inconsistencies in the account could be detected and corrected on the basis of the timeline, and missing facts filled in. In addition, questions about the timeline focused the officer's attention on events from more than a single time perspective, an approach having demonstrated utility for obtaining accurate eyewitness testimony [24].

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Step 4 — Decision Point Identification: During the timeline construction, specific decisions were identified for further probing. In some cases the verbal cues marking a decision were obvious (e.g., "I had to decide whether it was safe enough to send my crews inside"), but this was not always the case. In other cases, it would be clear that an officer was taking one of several possible courses of action or was making a judgment that affected the outcome, but there was no clear indication that the officer saw himself as "making a decision" at this point. A decision point was probed if the officer would agree that other reasonable courses of action were possible or that another officer (perhaps one with less or greater expertise) might have chosen differently.

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Step 5 — Decision Point Probing: Different studies have used different probes, depending on the objectives of the projects. Table I summarizes the probe types that have been routinely used.

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TABLE I
CRITICAL DECISION INTERVIEW PROBES

Probe Type	Probe Content
Cues	What were you seeing, hearing, smelling ...?
Knowledge	What information did you use in making this decision, and how was it obtained?
Analogues	Were you reminded of any previous experience?
Goals	What were your specific goals at this time?
Options	What other courses of action were considered by or available to you?
Basis	How was this option selected/other options rejected? What rule was being followed?
Experience	What specific training or experience was necessary or helpful in making this decision?
Aiding	If the decision was not the best, what training, knowledge, or information could have helped?
Time Pressure	How much time pressure was involved in making this decision? (Scales varied.)
Situation Assessment	Imagine that you were asked to describe the situation to a relief officer at this point, how would you summarize the situation?
Hypotheticals	If a key feature of the situation had been different, what difference would it have made in your decision?

Referências

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